

HU Collection

Test of Mathematics for University Admission

Mock Set J Paper 2

Mathematical Reasoning

Time allowed: 75 minutes

Questions: 20 multiple choice

Calculators: not permitted

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Date written: 29 September 2026

Name: _____

Date sat: _____

Score: / 20

Instructions

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, “ A only if B ” means “if A then B ”, and “ A unless B ” means “if B is false, then A ”. “Must be true” means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion. In a question that asks for the first mistake in an argument, a line counts as a mistake if it asserts something false or draws a conclusion that does not follow from the lines before it.

[illegible][illegible]

Question 1

An argument is *valid* if its conclusion must be true whenever all of its premises are true. In each argument below, n stands for an unknown positive integer. Which of the arguments are valid?

- I** If n is divisible by 6, then n is divisible by 3. The number n is not divisible by 3. Therefore n is not divisible by 6.
- II** If n is divisible by 6, then n is divisible by 3. The number n is divisible by 3. Therefore n is divisible by 6.
- III** Either n is even or n^2 is odd. The number n^2 is even. Therefore n is even.
- IV** No perfect square leaves remainder 2 on division by 4. The number n leaves remainder 2 on division by 4. Therefore n is not a perfect square.
- V** If n is a prime greater than 2, then n is odd. If n is odd, then $n^2 + 1$ is even. The number $n^2 + 1$ is even. Therefore n is a prime greater than 2.

- A** I only
- B** I and IV only
- C** I, III and IV only
- D** I, II and IV only
- E** I, III, IV and V only
- F** III and IV only
- G** all five

Question 2

At a sixth-form college, each student takes some of the four modules Pure, Mechanics, Statistics and Decision. The following facts are known.

- (1) A student takes Mechanics only if they take Pure.
- (2) Nobody takes both Statistics and Mechanics unless they also take Decision.
- (3) Every student who takes Decision also takes Statistics.
- (4) Aisha takes Mechanics but not Decision.

Here “ A unless B ” means “if B is false, then A ”. Which of the following can be deduced from these facts?

- I** Aisha takes Pure.
 - II** Aisha does not take Statistics.
 - III** Every student who takes both Mechanics and Decision also takes both Pure and Statistics.
 - IV** Every student who takes Decision takes Pure.
 - V** No student takes all four modules.
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- A** I only
 - B** I and II only
 - C** I, II and IV only
 - D** I, II and III only
 - E** I, II, III and V only
 - F** II and III only
 - G** I, III and IV only

Question 3

In this question a *geometric sequence* is one of the form $a, ar, ar^2, \dots$ for real numbers a and r . How many of the following statements about sequences of real numbers are true?

- I If a sequence is both arithmetic and geometric, then it is constant.
 - II If every term of a geometric sequence is positive, then the logarithms of its terms form an arithmetic sequence.
 - III If $u_1, u_2, u_3, \dots$ is arithmetic, then $2^{u_1}, 2^{u_2}, 2^{u_3}, \dots$ is geometric.
 - IV If $u_1, u_2, \dots$ and $v_1, v_2, \dots$ are both geometric, then $u_1 + v_1, u_2 + v_2, \dots$ is geometric.
 - V If $u_1, u_2, \dots$ is arithmetic and $u_1^2, u_2^2, \dots$ is also arithmetic, then $u_1, u_2, \dots$ is constant.
 - VI If the sum of the first n terms is $n^2 + n + 1$ for every $n \geq 1$, then the sequence is arithmetic.
- A 0
B 1
C 2
D 3
E 4
F 5

Question 4

A student attempts to prove the following claim.

Claim: if f is a polynomial with $\int_0^2 f(x) dx = 0$, then $f(1) = 0$.

- (1) The substitution $u = 2 - x$ shows that $\int_0^2 f(2 - x) dx = \int_0^2 f(u) du$.
- (2) Hence $\int_0^2 (f(x) + f(2 - x)) dx = 2 \int_0^2 f(x) dx = 0$.
- (3) Let $h(x) = f(x) + f(2 - x)$. Then $h(2 - x) = h(x)$ for every x , so the graph of h is symmetric about the line $x = 1$.
- (4) Since h is continuous and $\int_0^2 h(x) dx = 0$, there is some c in $[0, 2]$ with $h(c) = 0$.
- (5) By the symmetry in (3), also $h(2 - c) = 0$; so the zeros of h in $[0, 2]$ come in pairs c and $2 - c$, except that $x = 1$ is paired with itself.
- (6) A continuous function whose integral over an interval is 0 has an odd number of zeros in that interval, so $x = 1$ must be a zero of h .
- (7) Therefore $2f(1) = h(1) = 0$, and $f(1) = 0$.

Which one of the following is correct?

- A The first mistake is in line (1), and the claim is false.
B The first mistake is in line (3), and the claim is false.
C The first mistake is in line (4), and the claim is true.
D The first mistake is in line (5), and the claim is false.
E The first mistake is in line (6), and the claim is true.
F The first mistake is in line (6), and the claim is false.
G The argument is valid, and the claim is true.

Question 5

$\mathcal{F}$ is a collection of functions, each defined for all real numbers. Consider the statement “Every function in $\mathcal{F}$ that is increasing is bounded above, unless it is a polynomial.”

Here “ A unless B ” means “if B is false, then A ”. Which one of the following is the negation of the statement?

- A** There is a function in $\mathcal{F}$ that is increasing, is not a polynomial, and is not bounded above.
- B** There is a function in $\mathcal{F}$ that is increasing, is a polynomial, and is not bounded above.
- C** Every function in $\mathcal{F}$ that is increasing and not bounded above is a polynomial.
- D** There is a function in $\mathcal{F}$ that is not increasing, is not a polynomial, and is bounded above.
- E** Every function in $\mathcal{F}$ that is increasing is not bounded above, unless it is a polynomial.
- F** There is a function in $\mathcal{F}$ that is increasing and not bounded above, and every such function is a polynomial.
- G** No function in $\mathcal{F}$ is increasing.

Question 6

Consider the following statement about polynomials f and g .

If $f(x) \leq g(x)$ for every x with $0 \leq x \leq 1$, then $\int_0^1 (f(x))^2 dx \leq \int_0^1 (g(x))^2 dx$.

How many of the following pairs are counterexamples to the statement?

- (i) $f(x) = x$ and $g(x) = 1$
- (ii) $f(x) = -1$ and $g(x) = 0$
- (iii) $f(x) = x - 1$ and $g(x) = x$
- (iv) $f(x) = 2x - 2$ and $g(x) = x$
- (v) $f(x) = x^2$ and $g(x) = x$
- (vi) $f(x) = -x$ and $g(x) = x^2$
- (vii) $f(x) = x - 1$ and $g(x) = 1 - x$

- A** 2
- B** 3
- C** 4
- D** 5
- E** 6
- F** 7

Question 7

Consider the following statement, in which f is a polynomial.

If a straight line meets the curve $y = f(x)$ at exactly one point, then the line is the tangent to the curve at that point.

Which of the following are counterexamples to the statement?

- I The curve $y = x^2$ and the line $x = 1$.
 - II The curve $y = x^3$ and the line $y = 0$.
 - III The curve $y = x^3$ and the line $y = -x$.
 - IV The curve $y = x^2$ and the line $y = 2x - 1$.
 - V The curve $y = x^3 - 3x$ and the line $y = 9x + 16$.
 - VI The curve $y = x^4 + x$ and the line $y = x$.
- A I only
 - B III only
 - C I, III and V only
 - D III and V only
 - E I and III only
 - F I, III and VI only
 - G none of them

Question 8

For a polynomial f , consider the three conditions

- (1) $f(1) = 0$;
- (2) $\int_0^2 f(x) dx = 0$;
- (3) $f(1+t) = -f(1-t)$ for every real t .

Which one of the following is correct?

- A (1) alone is sufficient for (2).
- B (2) alone is sufficient for (1).
- C (1) and (2) together are necessary and sufficient for (3).
- D (1) and (2) together are sufficient but not necessary for (3).
- E (3) is sufficient for (1) but not for (2).
- F (1) and (2) together are necessary but not sufficient for (3).

Question 9

Three arguments are offered for the following claim, in which $g(x) = x^3 + x - 1$.

Claim: the equation $g(x) = 0$ has exactly one real solution.

- I $g(0) = -1 < 0$ and $g(1) = 1 > 0$, so g has a root between 0 and 1. Also $g'(x) = 3x^2 + 1 > 0$ for every x , so g is increasing and cannot have a second root.
- II The equation says $x^3 = 1 - x$. The graph of $y = x^3$ is increasing and the graph of $y = 1 - x$ is decreasing, so they meet at most once. They do meet, because at $x = 0$ the first graph is below the second and at $x = 1$ it is above.
- III $g(0) < 0 < g(1)$ gives a root r with $0 < r < 1$. If there were a second root s , then $r^3 + r = s^3 + s$, so $(r - s)(r^2 + rs + s^2 + 1) = 0$, and since $r \neq s$ this gives $r^2 + rs + s^2 = -1$. But r and s are both positive, being roots between 0 and 1, so $r^2 + rs + s^2 > 0$: a contradiction.

Which of these arguments are complete and correct proofs of the claim?

- A I and II only
- B I only
- C II only
- D I and III only
- E II and III only
- F I, II and III
- G none of them

Question 10

A student attempts to prove the following claim.

Claim: $x^4 - 4x + 3 \geq 0$ for every real number x .

- (1) If $x \leq 0$, then $-4x \geq 0$, so $x^4 - 4x + 3 \geq 3 > 0$.
- (2) If $0 < x \leq \frac{3}{4}$, then $3 - 4x \geq 0$, so $x^4 - 4x + 3 > 0$.
- (3) If $x \geq 2$, then $x^3 \geq 8$, so $x^4 \geq 8x$ and $x^4 - 4x + 3 \geq 4x + 3 > 0$.
- (4) If $\frac{3}{4} < x < 2$, then $x^4 - 4x + 3$ is a continuous function of x that is positive at $x = \frac{3}{4}$ and at $x = 2$, so it is positive for every x in between.
- (5) Hence $x^4 - 4x + 3 \geq 0$ for every real number x .

Which one of the following is correct?

- A The first mistake is in line (1), and the claim is false.
- B The first mistake is in line (2), and the claim is true.
- C The first mistake is in line (3), and the claim is true.
- D The first mistake is in line (4), and the claim is true.
- E The first mistake is in line (4), and the claim is false.
- F The first mistake is in line (5), and the claim is true.
- G The argument is valid, and the claim is true.

Question 11

The series $a + ar + ar^2 + ar^3 + \dots$, whose n th term is ar^{n-1} , has real first term a and real common ratio r . The condition

$$|r| < 1$$

is _____ for “the series converges and its sum is $2a$ ”.

- A necessary but not sufficient
- B sufficient but not necessary
- C necessary and sufficient
- D neither necessary nor sufficient

Question 12

For each real number k , let $f_k(x) = x^2 - kx$. How many of the following statements are true?

- (i) For every real k there is a real x with $f_k(x) < 0$.
- (ii) There is a real x such that $f_k(x) \leq 0$ for every real k .
- (iii) For every real x there is a real k with $f_k(x) < 0$.
- (iv) There is a real k such that $f_k(x) \geq -1$ for every real x .
- (v) For every real k there is a real x with $f_k(x) = f_k(x+1)$.
- (vi) There is a real x such that $f_k(x) \geq f_k(x+1)$ for every real k .

- A 1
- B 2
- C 3
- D 4
- E 5
- F 6

Question 13

Let a and b be positive real numbers, with $a \neq 1$. Which one of the following conditions is necessary and sufficient for $\log_a b > 1$?

- A $b > a$
- B $b > a > 1$
- C $(b-1)(a-1) > 0$
- D $(b-a)(b-1) > 0$
- E $b > a$ or $b < 1$
- F $(b-a)(a-1) > 0$

Question 14

Let f be a polynomial. Which of the following conditions are *sufficient* for $\int_0^1 f(x) dx > 0$?

- I $f(0) > 0$ and $f(1) > 0$.
 - II f has degree at most 1, and $f(\frac{1}{2}) > 0$.
 - III $f(x) > -1$ for every x in $[0, 1]$, and $f(x) > 1$ for every x in $[\frac{1}{4}, \frac{3}{4}]$.
 - IV f is increasing on $[0, 1]$, and $f(\frac{1}{2}) > 0$.
 - V $f(x) > 0$ for every x in $(0, 1)$ except $x = \frac{1}{2}$, and $f(\frac{1}{2}) = 0$.
- A II only
 - B II and III only
 - C II and V only
 - D II, IV and V only
 - E III and V only
 - F I, II, III and V only
 - G II, III and V only

Question 15

Which of the following conditions on real numbers x and y are equivalent to $|x| + |y| \leq 1$?

- I $|x + y| \leq 1$ and $|x - y| \leq 1$
 - II $|x| \leq 1$ and $|y| \leq 1$
 - III $x^2 + y^2 \leq 1$
 - IV $|x + y| + |x - y| \leq 2$
 - V $(|x| + |y|)^3 \leq |x| + |y|$
- A I only
 - B I and V only
 - C I, IV and V only
 - D II and IV only
 - E I, III and V only
 - F V only

Question 16

Let f be a polynomial that is not constant, and let a be a real number. Say that f has a local maximum at a if $f(x) \leq f(a)$ for every x in some interval around a , and similarly for a local minimum. Consider the conditions

$P: \quad f'(a) = 0;$

$Q: \quad f$ has a local maximum or a local minimum at a ;

$R: \quad f'(a) = 0$ and $f''(a) \neq 0$;

$S: \quad f'(x)$ has one sign for all x just less than a , and the opposite sign for all x just greater than a .

Which one of the following is correct?

- A** P , Q and S are equivalent, and R implies each of them.
- B** R and Q are equivalent, and each of them implies P .
- C** S implies R , and R implies P .
- D** P implies Q , and Q implies S .
- E** Q and S are equivalent and each implies P ; R implies Q ; but P does not imply Q , and Q does not imply R .
- F** Q implies S , but S does not imply Q .

Question 17

Call a function f , defined for every real number, a *crossing function* if for every real number c there are real numbers a and b with $a < b$ and $f(a) < c < f(b)$. How many of the following are crossing functions?

$$x^3 - 3x, \quad 1 - x, \quad x^2, \quad x + \sin x, \quad x \sin x, \quad \frac{x}{1 + |x|}$$

- A** 1
- B** 2
- C** 3
- D** 4
- E** 5
- F** 6

Question 18

For a real number x , consider the three statements

P : $x^2 < 4$ only if $x > 1$.

Q : $x > 1$ unless $x < -2$.

R : $x^2 > 1$ if $x < 0$.

Here “ A only if B ” means “if A then B ”, and “ A unless B ” means “if B is false, then A ”. For which values of x are exactly two of the statements P , Q and R true?

- A** no value of x
- B** $x = -2$ only
- C** $x \leq -2$
- D** $x = -2$ and $x = -1$ only
- E** $-2 \leq x < -1$
- F** $x < -2$ or $1 < x < 2$
- G** $-1 < x < 0$

Question 19

For a real number k , consider the three statements

- (1) for every x with $0 \leq x \leq 1$ there is a y with $0 \leq y \leq 1$ such that $x^2 + y^2 = k$;
- (2) there is an x with $0 \leq x \leq 1$ such that $x^2 + y^2 \leq k$ for every y with $0 \leq y \leq 1$;
- (3) for every x with $0 \leq x \leq 1$ there is a y with $0 \leq y \leq 1$ such that $x^2 + y^2 \geq k$.

Let $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ be the sets of real numbers k for which statements (1), (2) and (3), respectively, are true. Which one of the following is correct?

- A** $\mathcal{A} = \{1\}$, $\mathcal{B} = \{k : k \geq 1\}$, $\mathcal{C} = \{k : k \leq 1\}$
- B** $\mathcal{A} = \{k : 1 \leq k \leq 2\}$, $\mathcal{B} = \{k : k \geq 1\}$, $\mathcal{C} = \{k : k \leq 1\}$
- C** $\mathcal{A} = \{1\}$, $\mathcal{B} = \{k : k \geq 2\}$, $\mathcal{C} = \{k : k \leq 2\}$
- D** $\mathcal{A} = \{k : 0 \leq k \leq 2\}$, $\mathcal{B} = \{k : k \geq 1\}$, $\mathcal{C} = \{k : k \leq 1\}$
- E** $\mathcal{A} = \{1\}$, $\mathcal{B} = \{k : k \geq 1\}$, $\mathcal{C} = \{k : k \leq 2\}$
- F** $\mathcal{A}$ is empty, $\mathcal{B} = \{k : k \geq 2\}$, $\mathcal{C} = \{k : k \leq 1\}$

Question 20

For which of the following sets of conditions does there exist a polynomial f satisfying every condition in the set?

- I** $f(0) = 0$, $f(1) = 1$, and $\int_0^1 f(x) dx = 0$.
 - II** $f(0) = f(1) = 0$, $f(x) \geq 0$ for $0 \leq x \leq 1$, and $\int_0^1 f(x) dx = 1$.
 - III** $f(0) = 0$, $f'(x) \leq 1$ for $0 \leq x \leq 1$, and $\int_0^1 f(x) dx = \frac{1}{2}$.
 - IV** $f(0) = 0$, $f(1) = 1$, $f'(x) \geq 0$ for $0 \leq x \leq 1$, and $\int_0^1 f(x) dx = \frac{1}{100}$.
 - V** $f(0) = 0$, $f(1) = 1$, $f'(x) \geq 0$ for $0 \leq x \leq 1$, and $\int_0^1 f(x) dx = 1$.
- A** I and II only
 - B** I, II and IV only
 - C** II and IV only
 - D** I, II, IV and V only
 - E** I, IV and V only
 - F** all five
 - G** I, II, III and IV only